

ICML 2026

DEER: A Benchmark for Evaluating Deep Research Agents on Expert Report Generation

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Motivation

A Long, Cited Report Is Not Necessarily a Good Report

Deep research agents can now generate

- ✓ Structured sections
- ✓ Fluent writing
- ✓ Professional tone
- ✓ Many citations

But expert users need more than plausibility:

- 1 ⚠ Did it address the real goal?
- 2 ⚠ Did it cover the right scope?
- 3 ⚠ Is the analysis complete and coherent?
- 4 ⚠ Are key claims supported?

Expert Report

Global Battery Supply Chain Risk Assessment

May 20, 2025

1. EXECUTIVE SUMMARY

1 This report provides a broad overview of battery supply chain risks and market conditions through 2030.

The battery industry is entering a decade of rapid growth, driven by electrification across transportation, grid storage, and consumer electronics. However, this expansion is accompanied by material risks across critical minerals, manufacturing capacity, logistics, and geopolitical uncertainty. We summarize major risk categories and potential market impacts for manufacturers and investors.

2. METHODOLOGY

2 We evaluate risks across critical minerals, refining capacity, cell manufacturing, and long-term demand growth.

The analysis combines public market reports, company disclosures, trade data, and expert judgment to score major risks by likelihood and impact from 2025 to 2030.

3. KEY FINDINGS / ANALYSIS

Risk Category	Description	Likelihood (2025-2030)	Impact	Risk Level
Critical mineral concentration	Limited supply exposure for lithium, nickel, cobalt, and graphite.	High	High	High
Commodity price volatility	Price swings driven by demand surges, speculation, and constrained supply.	High	Medium	High
Manufacturing bottlenecks	Capacity constraints and specialized shortages delay production.	Medium	High	High
Geopolitical policy risk	Trade restrictions, export controls, and policy changes increase uncertainty.	Medium	High	Medium

4. SCENARIO OUTLOOK

Scenario	Description	Cost Impact (2025-2030)
Upside case	Stable supply and faster recycling scale-up	-5% to -10%
Base case	Current trends continue	-5% to +10%
Downside case	Trade barriers and material shortages intensify	+15% to +30%

5. RECOMMENDATIONS

- Diversify sourcing for critical minerals and battery components.
- Negotiate long-term offtake agreements with qualified suppliers.
- Build inventory buffers for high-risk materials.

- Expand recycling partnerships to reduce primary mineral dependency.
- Monitor policy changes affecting tariffs, subsidies, and export controls.

KEY TAKEAWAY

4 Strategic stockpiling is the single most effective near-term intervention.

REFERENCES

[1] IEA, Global EV Outlook 2024.
[2] BloombergNEF, Lithium-ion Battery Supply Chain Outlook 2024.
[3] McKinsey & Company, Battery Supply Chain Resilience in a Volatile World.

Projected Battery Demand by Region (GWh)

Motivation

Current Evaluation Struggles to Capture Expert-Level Report Quality

Existing benchmarks face four core challenges:

1 What should be evaluated?

→ Lack of systematic expert criteria

2 Where did the system fail?

→ Single or coarse scores hide specific weaknesses

3 What would an expert notice?

→ Generic judges miss task-specific requirements

4 Are claims truly supported?

→ Evidence verification does not cover the whole report

DEER

Making expert report evaluation systematic, interpretable, and evidence-grounded

DEER addresses the four evaluation gaps with:

Expert report taxonomy

→ Defines what to evaluate

(7 dimensions, 25 subdimensions)

Task-specific expert guidance

→ Helps judges catch issues requiring expert judgment

(expert-written guidance for each task)

Fine-grained dimensions and rubrics

→ Identifies where systems fail

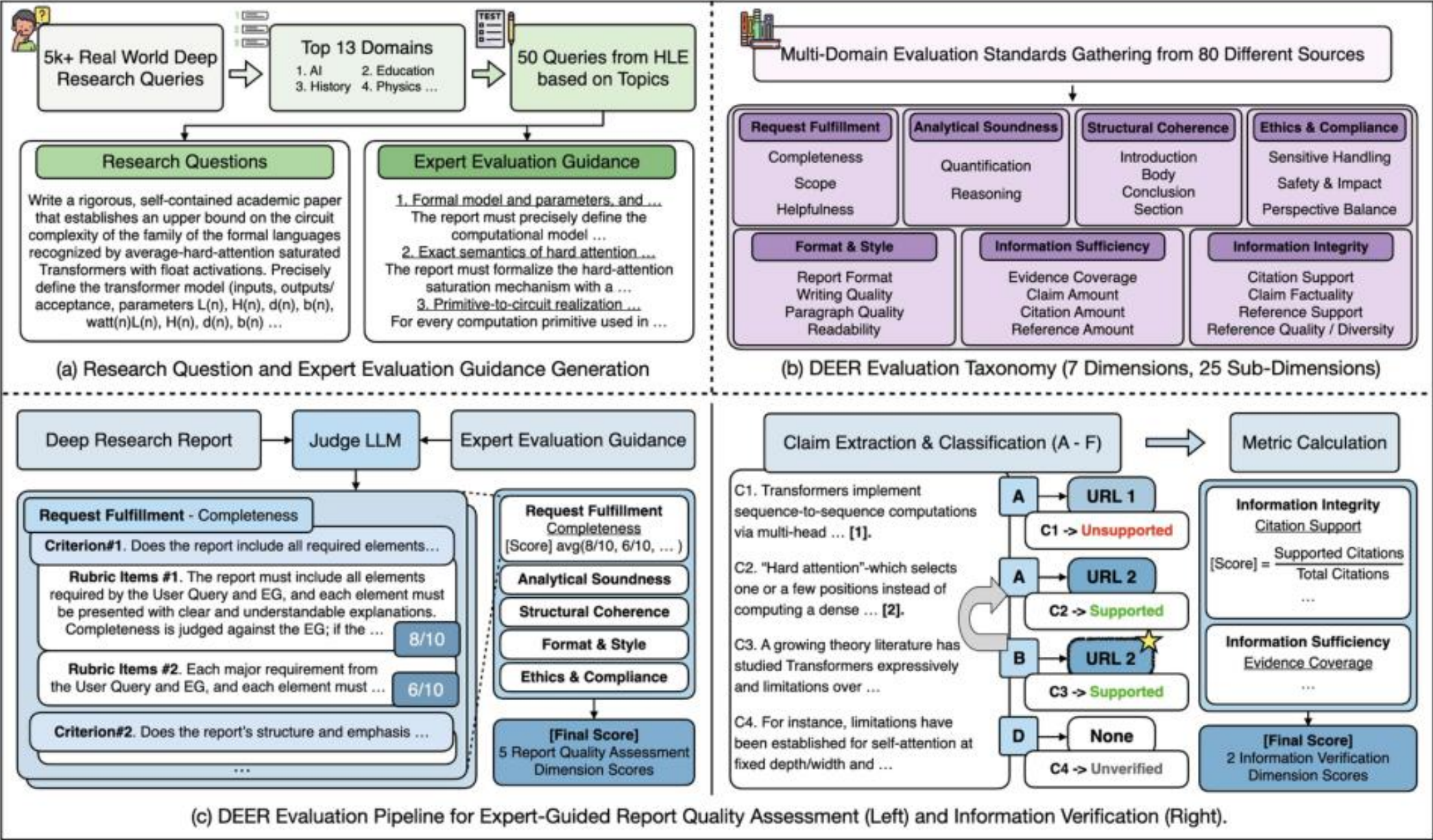
(101 rubric items)

Report-wide verification

→ Assesses full-report evidence support

(claim extraction + verification + metrics)

Overview



Information Verification

From report claims to report-level information metrics

1 Claim Extraction & Classification

Extract atomic claims and classify their evidence links (A–F) types

Example Report Snippet

- S1** Multi-junction solar cells achieved >45% efficiency in labs [1].
- S2** This improvement comes from layering different semiconductor materials.
- S3** These panels can withstand Category 4 hurricanes.

Extracted Atomic Claims

- A** “achieved >45% efficiency” [1]
→ Explicitly cited claim
- B/C** “improvement comes from layering different materials”
→ Implicit claim
- F** “withstand Category 4 hurricanes”
→ No recoverable source

A – explicit cited **B/C** – Implicit cited **D/E** – no ext. verification needed **F** – no recoverable source

2 Back-tracking

Recover evidence links for implicit claims

Recovering Evidence Links

- A** S1: explicit citation
→ directly linked to [1]
- B/C** S2: no citation found
→ back-track to S1 and inherit [1]
- F** S3: no citation found
→ no source found in earlier context (no source)

→ Direct link - - - → Back-tracked link

3 Verification + Signal Extraction

From verified claims and linked references, derive evidence signals

Verification (support check)

- A** Verify claim against [1] ✓
- B/C** Verify claim against [1] ✓
- F** Not verifiable ✗

Evidence Signals

From verified claims and linked references, derive evidence signals:

- ✓ Claim Factuality
- 🔗 Citation Support
- 📖 Reference Support
- 🔄 Reference Reproducibility
- 🛡️ Reference Reliability
- 👥 Reference Diversity

Derived from verified claims and linked references.

4 Metric Output

Aggregate evidence signals into report-level metrics

Information Integrity

- 🛡️ Claim Factuality
- 🔗 Citation Support
- 📖 Reference Support
- 🌟 Reference Quality
- 🔄 Reproducibility
- 🛡️ Reliability
- 👥 Reference Diversity

Information Sufficiency

- 🕒 Evidence Coverage
- 📄 Information Amount
- 🔗 Citation Amount
- 📖 Reference Amount

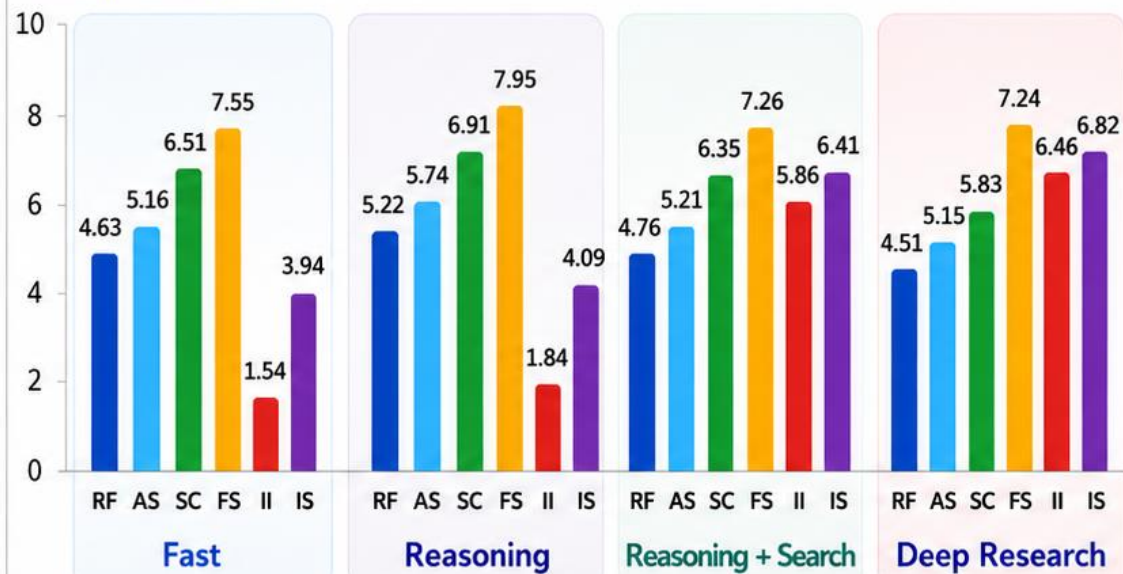


DEER turns claim-level evidence checks into report-level information scores.

Experiments

System-level trend

Average Score (1–10)



Expert Quality:
RF + AS



Presentation:
SC + FS



Evidence Use:
II + IS



Search/deep research improves evidence grounding,
but expert-level quality remains challenging.

RF Request Fulfillment AS Analytical Soundness SC Structural Coherence
FS Format & Style II Information Integrity IS Information Sufficiency

Fine-grained diagnostic scores

Selected subdimensions with an example rubric-item drill-down

1 Request Fulfillment



+ more subdimensions

2 Analytical Soundness



+ more subdimensions

3 Information Sufficiency



+ more subdimensions

4 Information Integrity



+ more subdimensions

Example rubric-item drill-down

Completeness

Overall score

5.1

Required elements
included & explained 4.8

Major requirements
sufficiently developed 5.3



Shown: selected subdimensions and example rubric items

Full DEER: 7 dimensions, 25 subdimensions, 101 rubric items

Thank you